ECONOMIC RESEARCH NOTE / 2026.10

THE BRITISH MULTIPLIER ECONOMY

A Neo-Keynesian guide to credit, industrial linkages and productive investment

The policy question is not simply how much money enters the economy. It is how far that money travels through British supply chains, household incomes, exports and productivity - and how much new productive capacity remains when the first round of spending has ended.

SCOTTISH FREE BANK | RESEARCH & STRATEGY

Prepared for strategic discussion. The sector chart is an explicitly analytical index, not a set of official UK multiplier estimates.

01 / EXECUTIVE VIEW

Finance should multiply productive capacity

A Neo-Keynesian view begins with aggregate demand: when one firm invests, suppliers receive orders, workers receive income and some of that income is spent again. But in a modern economy the size of this chain depends on the composition of the investment, the domestic content of its supply chain, available capacity and the response of monetary policy.

For a Scottish free bank, the practical question is how to distinguish credit that merely changes ownership of existing assets from credit that finances additional production, infrastructure, export capacity and productivity. The objective is not maximum lending in isolation; it is a better conversion of finance into real output and sustainable cash flows.

Five conclusions

Money creation is not wealth creation.

Commercial banks normally create deposit money when they lend. What matters for prosperity is the productive use of the financing, not the deposit entry alone. [1]

There is more than one multiplier.

The expenditure multiplier follows income and re-spending; the input-output multiplier traces direct and indirect supplier demand; the productivity effect captures benefits to output over time.

Industrial linkages matter.

Manufacturing, energy networks, processing and infrastructure can create demand across several domestic industries. Their multiplier is stronger when viable British suppliers and downstream users exist.

Digital infrastructure is a conditional case.

Data centres create construction and electrical demand, but imported hardware and power constraints can reduce the immediate domestic effect. Their long-run case depends on the productivity gains they enable elsewhere.

Prices and capacity discipline the outcome.

If labour, grid connections or components are scarce, extra spending may raise prices or imports instead of real domestic output. Supply-side investment must accompany demand.

> **THE CENTRAL THESIS**
>
> Britain needs a multiplier of productive capacity: investment that raises current demand while building the ability to produce more tomorrow.

02 / ECONOMIC FRAMEWORK

Three multipliers, not one

The word “multiplier” is used for different economic mechanisms. Combining them without distinction can make a financing decision look more certain than the evidence supports.

1. Money creation and bank credit

When a commercial bank makes a loan, it normally credits the borrower with a deposit and records a corresponding loan asset. Repayment reduces both the outstanding loan and the deposit money associated with the transaction. This is not a fixed-reserve process in which a predetermined quantity of central-bank money is mechanically multiplied. Bank lending is conditioned by monetary policy, capital, risk, borrower demand and expected returns. [1]

The bank therefore chooses which future cash flows to finance. A loan for new productive equipment has a different real-economy profile from a loan that only facilitates a transfer of an existing asset.

2. The Keynesian expenditure multiplier

An initial injection becomes income for someone else. Recipients spend a share of that income, supporting further production and income. Saving, taxation and imports interrupt the chain; a shortage of capacity can turn some of the additional nominal demand into higher prices.

SIMPLIFIED OPEN-ECONOMY FORMULA

$$k = 1 / [1 - c(1 - t) + m]$$

c = marginal propensity to consume; t = effective marginal tax rate; m = marginal propensity to import.

Illustration, not a UK estimate: if c = 0.8, t = 0.25 and m = 0.2, the model gives k = 1.67. A £100m injection would imply approximately £167m of total expenditure/output in the simplified model, subject to its assumptions. It is not a forecast and should not be read as £167m of extra UK value added.

3. The input-output and productivity multipliers

Input-output analysis maps the direct order and the subsequent orders to suppliers. ONS Type 1 multipliers include direct and indirect supply-chain effects; they do not include the induced household-spending effect sometimes described as Type 2. ONS also cautions that these estimates assume the change in supply can be met and prices remain unaffected. [2][3][4]

A further productivity channel may extend beyond the conventional production multiplier: a grid upgrade, machine tool or data centre can raise the output of other firms over years. It is economically important, but must be evaluated separately from an input-output ratio.

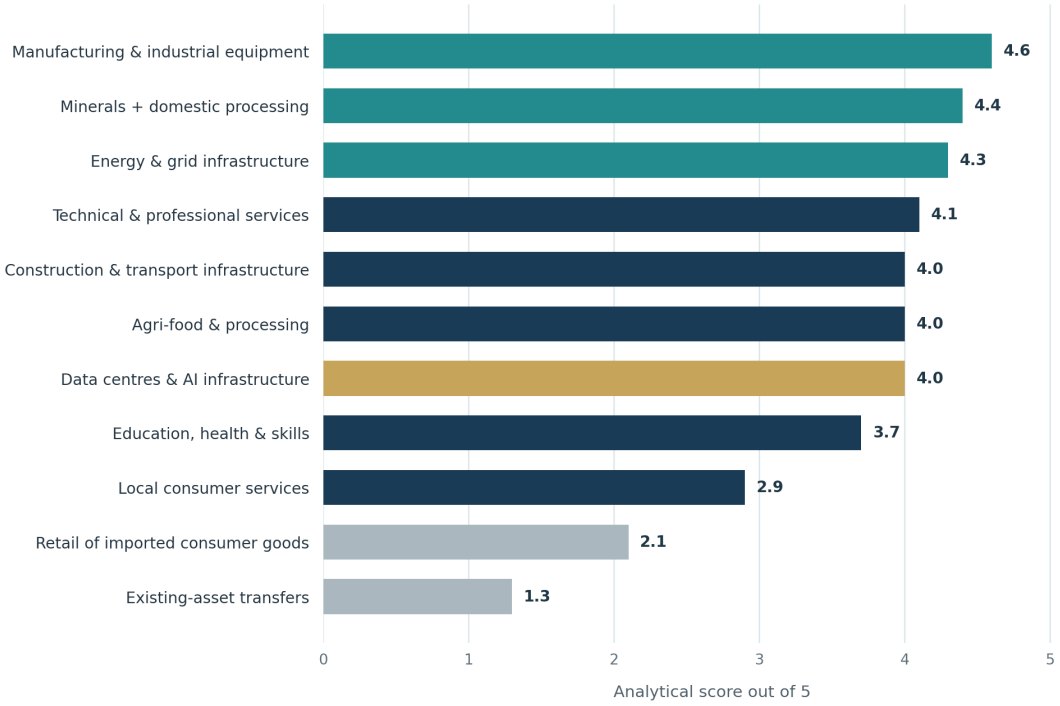
03 / SECTOR COMPARISON

Where might the multiplier be strongest?

The index below is a bank-authored screening framework for discussion. It asks which activities are most likely to connect domestic supply chains, circulate income, earn export revenue or replace imports, and increase productivity elsewhere. It is not an official ONS multiplier and is not a ratio of GDP generated per pound spent.

SECTORAL MULTIPLIER POTENTIAL INDEX

Scottish Free Bank analyst framework | 12 October 2026 | Score, not an official £/£ multiplier



Method: a weighted assessment out of 5. Domestic supplier linkages (30%); induced income circulation (20%); export capability or competitive import replacement (20%); long-run productivity spillovers (30%). Rounded scores are analyst judgements for comparative screening, not measured statistical estimates. ONS official input-output and employment multipliers should be used for formal impact appraisals. [2][3][4]

How to read the ranking

GROUP	INTERPRETATION
Higher potential	Manufacturing, mineral processing and energy networks combine supplier demand with useful capacity that can unlock further activity.
High but conditional	Data centres, AI and tradable professional services have substantial productivity or export potential, but results depend on domestic adoption, skills, energy and imported inputs.
Lower direct production effect	Imported-goods retail and transfers of existing assets can provide useful services or liquidity, but usually generate less incremental domestic production at the point of transaction.

The score should inform questions, not dictate a lending decision. A lower-scoring project may still be essential infrastructure; a high-scoring project may still be uneconomic or inflationary if its costs exceed the value it creates.

04 / INDUSTRIAL CHANNELS

Mining, manufacturing, energy and computing

Mining: extraction is the starting point, not the whole value chain

A mine creates demand for surveying, engineering, machinery, maintenance, water treatment, environmental services and freight. The multiplier can deepen if commercially competitive processing takes place in Britain and the material feeds domestic manufacturers. If ore or minerals are exported with little processing, or if equipment and profits largely flow abroad, the domestic effect is narrower. The right question is which parts of the chain Britain can supply competitively, not whether every stage must be local.

Manufacturing: a network of backward and forward linkages

A factory buys material, components, software, energy, logistics and professional services. Its products may then make other firms more productive. That combination gives manufacturing a potential multiplier through supplier demand, exports, learning and the use of machinery downstream. Automation may limit jobs per pound invested while still increasing productive capacity; employment alone is therefore an incomplete scorecard.

Energy and infrastructure: the complementarity effect

Reliable electricity, grid capacity, rail freight, ports and fibre networks are enabling assets. One investment raises the return on others: an industrial project may be commercially impossible without a substation or freight connection. The economic return therefore includes the productive activity made possible by the asset, as well as the construction contracts it initially generates. Planning delays and scarce engineers can erode these benefits.

Data centres and AI: a two-stage investment

The construction phase supports civils, electrical equipment, cooling, networks and site services, but imported servers and accelerators are a leakage from UK production. The operating facility may have a relatively small permanent workforce. Its broader value depends on whether British manufacturers, researchers, farms, financial firms and professional services use the computing capacity to improve output, quality or speed. Power availability and grid queues are core parts of the multiplier analysis.

Agri-food: productivity plus domestic processing

Farm technology and better storage can create demand for machinery, agronomy and logistics, reduce waste and improve output. Connecting farms to competitive British food processors, packaging, cold-chain and export channels can retain more value added. The goal is not autarky: imported inputs can be efficient, and domestic processing should be judged by productivity and market demand.

05 / INCOME CIRCULATION AND CONSTRAINTS

Services, household demand and the inflation boundary

Services should not be treated as one undifferentiated sector. A local cafe, a software exporter, a hospital and an engineering consultancy produce different combinations of income circulation, trade and productivity spillovers.

Local consumer services

Restaurants, retail, repairs and personal services can create a relatively visible induced-spending loop: wages and supplier payments are spent in nearby businesses. This can matter greatly for local employment and town-centre demand. Yet some spending displaces purchases elsewhere, some revenue buys imported goods, and capacity constraints can raise prices instead of increasing output.

Professional and digital services

Engineering, software, research, design, finance and specialist consulting can sell overseas and improve other firms' productivity. Their direct employment multiplier may differ from that of manufacturing, but knowledge diffusion and export earnings can make their wider economic value substantial. Human-capital investment in education, health and skills can also improve productive capacity over time.

Three leakages to track

LEAKAGE / CONSTRAINT	ECONOMIC INTERPRETATION
Imports	Imported inputs satisfy real needs but mean less of the initial expenditure is UK production. Evaluate domestic value added, not simply gross contract value.
Saving and retained profits	Not necessarily a loss: funds can support later investment. The question is whether saving is intermediated into productive projects or remains inactive / is used mainly for existing assets.
Tax and monetary response	Taxation changes disposable income; higher inflation pressure can prompt tighter monetary conditions, dampening private spending and investment elsewhere.

The inflation boundary

A demand injection has the greatest chance of raising real output when firms can respond by employing available resources, expanding shifts or commissioning investment. If grid connections, skilled workers, building materials or specialist components are fixed in the short run, the same injection may bid up wages and prices or increase imports. A Neo-Keynesian policy therefore pairs demand support with measures that expand supply and remove bottlenecks.

Regional multipliers also differ. Spending that leaves a Scottish region may still add to UK output; imported equipment is a national leakage. Regional supplier development can strengthen the local loop where commercially viable, but indiscriminate local-content requirements can increase costs and weaken the project itself.

06 / IMPLICATIONS FOR SCOTTISH FREE BANK

A productive-credit scorecard

A Neo-Keynesian lending framework should not allocate credit only by sector label. It should underwrite the project's economics and then map the economic pathways through which it may create value.

TEST	UNDERWRITING QUESTION
1 Cash-flow viability	Can the borrower service debt under conservative prices, volumes, interest rates and commissioning assumptions?
2 Incremental capacity	Does the project create new output, modernise productive assets or remove a binding constraint, rather than mainly changing ownership?
3 Domestic linkage	What proportion of expenditure becomes UK value added, and which local/regional suppliers are genuinely competitive?
4 Trade and strategic position	Can it earn export revenue, improve export competitiveness or displace imports at comparable quality and full cost?
5 Productivity externalities	Does it improve productivity for other firms through infrastructure, tools, data, skills or knowledge transfer?
6 Delivery and inflation risk	Are grid access, labour, planning, inputs and commissioning dates credible? Could project demand overwhelm near-term supply?

Policy implications

• Coordinate investment in productive assets and the infrastructure that makes them usable: power, grid, transport, ports, skills and housing.

• Use ONS input-output tables to estimate direct and indirect industry effects, then add a separate assessment of household-induced demand, productivity spillovers and price/capacity constraints.

• Prefer supply-chain development where it is competitive; do not equate national resilience with producing every component domestically.

• Distinguish lending against rising collateral values from credit that finances new productive output, while recognising that property, working capital and consumer finance can all meet legitimate needs.

Conclusion

The best multiplier is not a mythical fixed number attached to a sector. It is the durable connection between finance, additional demand, domestic value added and future productive capacity. Manufacturing can transmit orders through suppliers; energy and transport can unlock whole clusters; computing can lift productivity across industries; services and household spending can carry income through communities. Each mechanism has different leakages and risks. A bank that measures them separately can finance more of the projects that turn credit into productive wealth - without confusing the creation of money with the creation of value.

07 / METHODOLOGY AND REFERENCES

Evidence, definitions and limitations

This note combines standard Keynesian reasoning with current UK statistical frameworks. The sector chart is a transparent, analyst-scored comparison designed to structure questions for discussion. It is not a measured output multiplier, employment multiplier, forecast, credit rating or estimate of the return on any specific investment.

For quantitative appraisal, the starting point should be the ONS UK Input-Output Analytical Tables and Supply and Use Tables. Those data describe intermediate inputs, final demand, output and value added across the UK economy. ONS also publishes sectoral employment multiplier estimates. Type 1 input-output multipliers capture direct and indirect effects; they do not capture household-induced effects. They assume, among other things, that supply can meet extra demand and prices are unchanged, so they need careful use when capacity constraints are binding. [2][3][4]

Sources

[1] Bank of England (2014), “Money creation in the modern economy”, Quarterly Bulletin, Q1.
https://www.bankofengland.co.uk/quarterly-bulletin/2014/q1/money-creation-in-the-modern-economy

[2] Office for National Statistics (11 Dec 2025), “UK Input-output analytical tables: 2023”.
https://www.ons.gov.uk/releases/ukinputoutputanalyticaltables2023

[3] Office for National Statistics (19 Dec 2025), “Employment multipliers and effects in the UK: 2023”. Statistics are designated official statistics in development.
https://www.ons.gov.uk/releases/employmentmultipliersandeffectsintheuk2023

[4] Office for National Statistics (revised 1 Aug 2025), “Input-Output Analytical Tables: Quality and Methodology Information”.
https://www.ons.gov.uk/economy/nationalaccounts/supplyandusetables/methodologies/inputoutputanalyticaltablesqmi

[5] Office for National Statistics (5 Mar 2025; tool updated 11 Dec 2025), “How do different industries and products affect the UK economy?”
https://www.ons.gov.uk/economy/nationalaccounts/supplyandusetables/articles/howdodifferentindustriesandproductsaffecttheukeconomy/2025-03-05

Research note prepared for discussion by Scottish Free Bank. All sector scores and recommendations are analytical judgements. Formal project assessment requires current project-level costs, financing terms, supply-chain data, capacity analysis and a counterfactual.

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Credit for production. Capital for capacity. Multipliers for the real economy.